



EfficientFlow

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Efficient Equivariant Flow Policy Learning for Embodied AI

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Project page

efficientflow.github.io



Code

github.com/chang-jl/EfficientFlow



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1 Slow and data-hungry policies

Diffusion-style visuomotor policies learn complex action distributions from high-dimensional observations, but two limitations hold them back.

- **Data inefficiency:** they need large numbers of demonstrations.
- **Sampling inefficiency:** every action takes iterative denoising, 100 network evaluations (NFE) for Diffusion Policy [5] and EquiDiff [2].

EfficientFlow addresses both in one flow-based framework.

● **Equivariance for data efficiency.** With an isotropic prior and an equivariant velocity network (Theorem 1), if the observation rotates, the action naturally rotates. This yields a massive data advantage: a single demo effectively serves as training data for multiple rotations.

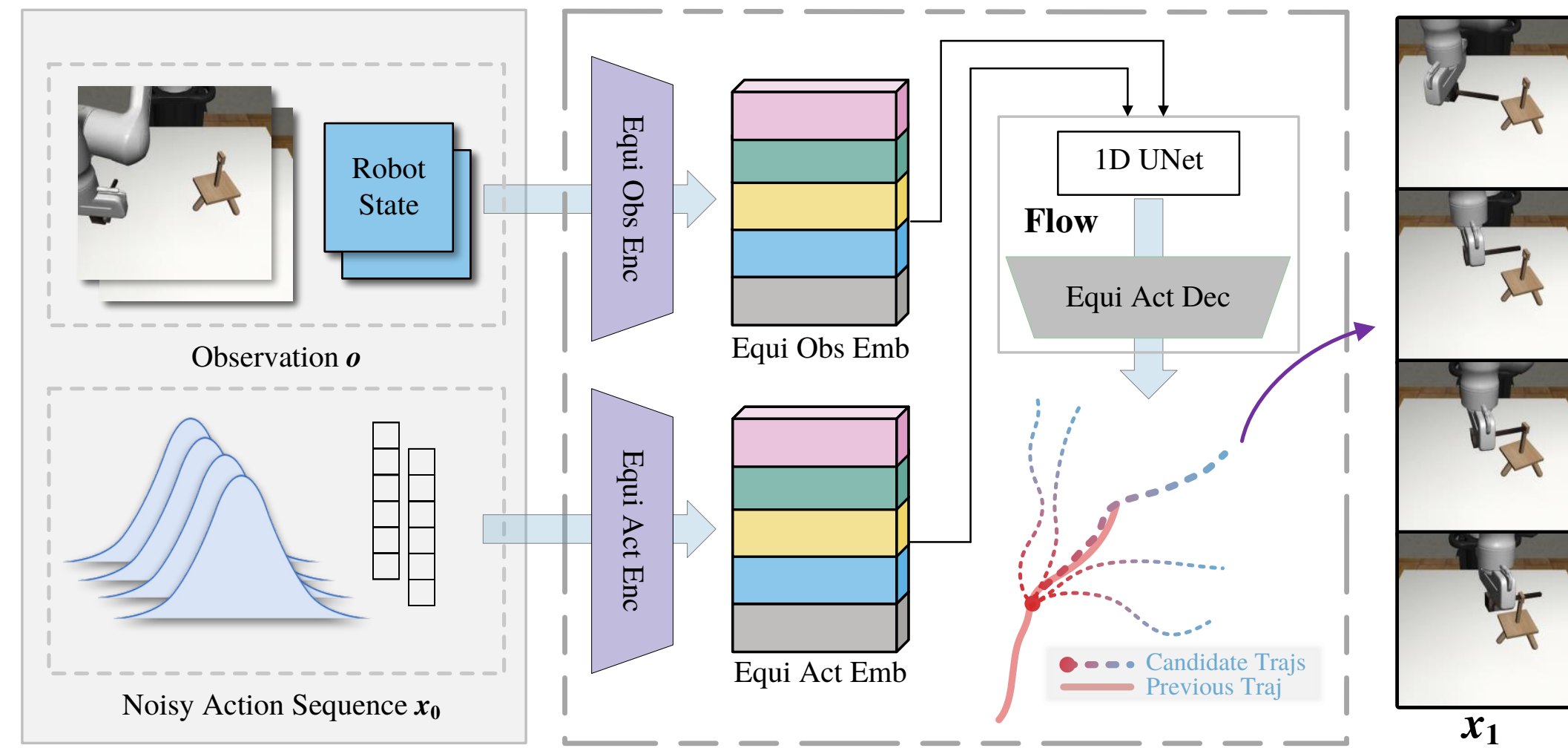
● **Acceleration regularization for fast sampling.** Penalizing the second time-derivative of the flow encourages smoother, low-curvature sampling trajectories. FABO makes the penalty trainable from conditional trajectories alone.

1NFE one network pass per action sequence

12.2ms per inference (81.8 Hz)

56.1× faster than EquiDiff (685.9 ms, 100 NFE)

98.4% of DP-C's 1000-demo success, from 200 demos (5 NFE)



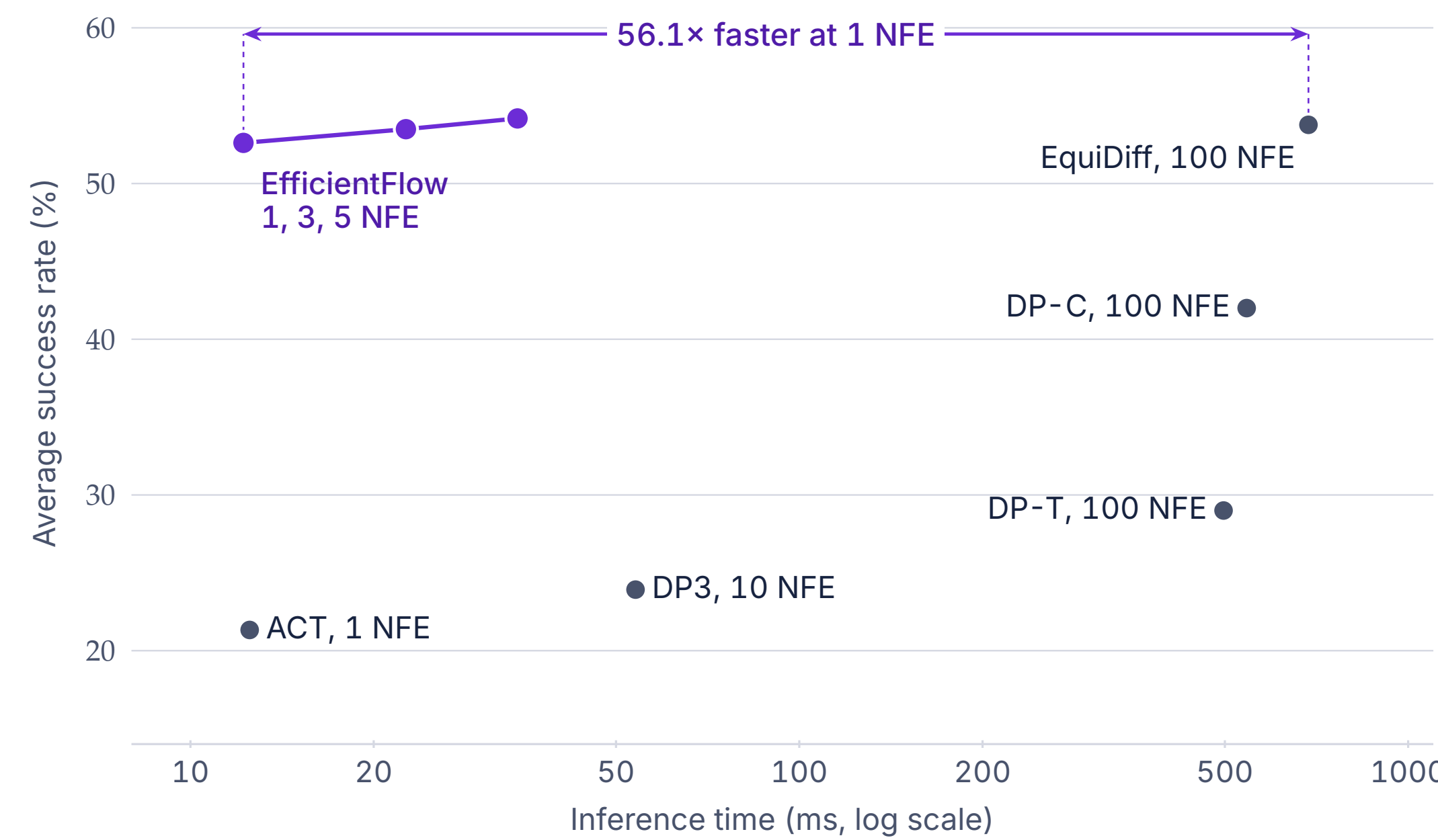
Pipeline. The policy reads the two latest observations o . The equivariant network transports five noise samples to candidate action trajectories in parallel. The candidate closest to the previous prediction on the overlapping steps is executed; every 10 cycles a random one is taken to keep the policy multi-modal. This selection-and-reset strategy is inspired by IMLE Policy [12].

Symmetry Encoders	cyclic group $C_n \subset \text{SO}(2)$ of planar rotations, built with escnn [13]
Velocity field	equivariant ResNet-18 (agent view), ResNet-18 (wrist camera)
Actions	1D U-Net (512, 1024, 2048), equivariant action encoder and decoder
Sampling	16-step sequences of absolute 6-DoF pose and gripper width; 8 executed Euler integration with 1, 3 or 5 steps

4 Experiments on MimicGen [4]

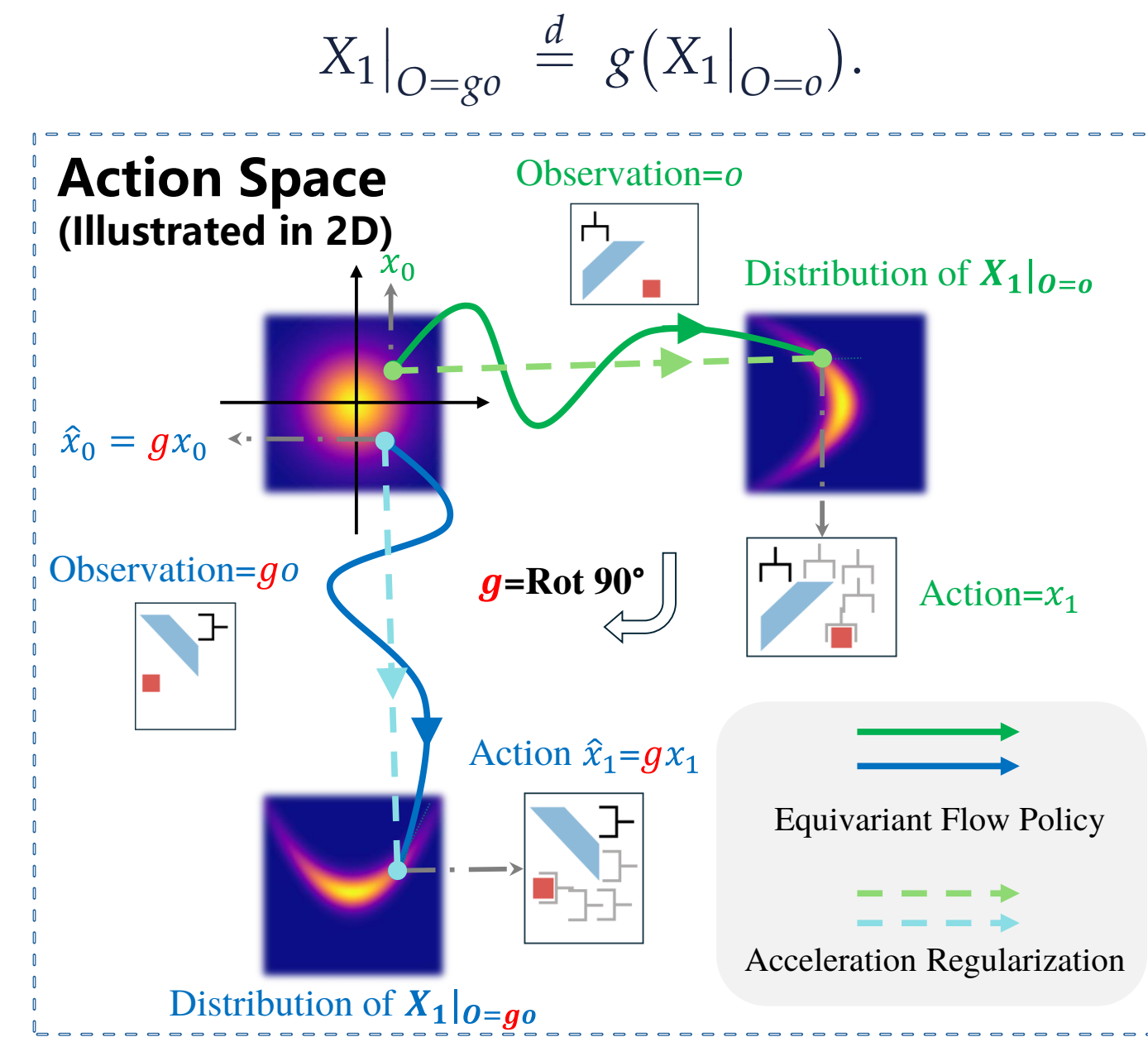
12 manipulation tasks, trained with 100, 200 and 1000 demonstrations, three seeds. RGB methods see the agent-view and wrist cameras and predict absolute poses. EfficientFlow runs with 1, 3 and 5 NFE; baseline success rates are those reported by EquiDiff [2].

Average success vs. inference time, 100 demonstrations



2 Equivariant flow policy

A symmetry of the observation should induce the same symmetry of the action distribution. For every g in the group G (here planar rotations),



Rotating the observation by g rotates the whole flow: the blue trajectory is the green one transformed by g . Dashed lines: with acceleration regularization.

Two ingredients are enough:

- an **isotropic distribution** for p_0 , e.g. Gaussian noise $X_0 \sim \mathcal{N}(0, I)$, the flow-matching default;
- an **equivariant velocity network**, $u_\theta(t, gx | go) = g u_\theta(t, x | o)$ for all $g \in G$.

Theorem 1. If p_0 is isotropic and u_θ is equivariant, the distribution induced by the flow ODE $\frac{dx_t}{dt} = u_\theta(t, x_t | o)$, $x_0 \sim p_0$, satisfies

$$X_t|_{O=g*o} \stackrel{d}{=} g(X_t|_{O=o}) \quad \text{for all } t \in [0, 1].$$

Proof idea. Let $\Phi_t(x_0 | o)$ be the flow map. $g \Phi_t(x_0 | o)$ and $\Phi_t(gx_0 | go)$ solve the same ODE (under observation go) from the same initial value, so they coincide by uniqueness. Since p_0 is isotropic, $gX_0 \stackrel{d}{=} X_0$, hence

$$g(X_t|_{O=o}) = \Phi_t(gX_0 | go) \stackrel{d}{=} \Phi_t(X_0 | go) = X_t|_{O=g*o}.$$

- The expert policy behind the demonstrations is *not* assumed to be equivariant, in contrast to EquiDiff [2].
- Actions transform under $\rho_1^3 \oplus (\rho_1 \oplus \rho_0) \oplus \rho_0$: 6D rotation [14], translation (xy and z), gripper width.

Method	NFE	Stack D1	Square D2	Threading D2	Stack Three D1	Coffee D2	3-Pc. Asm. D2	Hammer Cl. D1	Mug Cl. D1	Kitchen D1	Pick Place D0	Nut Asm. D0	Coffee Prep D1	Avg.
EfficientFlow	1	94	21	31	48	65	11	75	50	66	37	59	75	52.6
EfficientFlow	3	88	20	31	49	66	11	72	50	73	41	61	81	53.5
EfficientFlow	5	87	22	31	50	67	11	74	50	73	41	62	83	54.2
EquiDiff [2]	100	93	25	22	55	60	15	65	50	67	42	74	77	53.8
DP-C [5]	100	76	8	17	38	44	4	52	43	67	35	55	65	42.0
DP-T [5]	100	51	5	11	17	47	1	48	30	54	15	31	38	29.0
DP3 [8] (point cloud)	10	69	7	12	7	34	0	54	21	45	12	16	10	23.9
ACT [9]	1	35	6	10	6	19	0	38	23	37	7	42	32	21.3
Ablations of EfficientFlow: a component removed, or our formulation replaced by another one-step method on the same equivariant network														
No acceleration reg.		88	16	24	44	56	10	56	28	42	16.5	28	62	39.3
No equivariance		88	12	12	14	52	0	66	38	54	17.5	43	56	37.7
Consistency FM [6]		88	8	22	44	60	8	54	36	66	19	40	40	40.4
MP1 [7]		96	22	26	34	50	8	58	50	62	26	51	72	46.3

Success rate (%) with 100 demonstrations. Upper block: maximum over 50 evaluations during training, mean of three seeds; bold marks the best method per task. EfficientFlow outperforms EquiDiff on 7 of 12 tasks while using 1 to 5 network evaluations instead of 100.

3 Acceleration regularization and the FABO identity

Flow policies trained with conditional flow matching [1] alone perform poorly at low NFE, which suggests overly curved flow fields. We add an acceleration regularization term that encourages smoother, low-curvature trajectories,

$$\underbrace{\mathbb{E} \|u_\theta(t, x_t) - u(t, x_t | x_1)\|_2^2}_{\text{data term}} + \underbrace{\mathbb{E} \left[\lambda(t) \left\| \frac{d^2 x_t}{dt^2} \right\|_2^2 \right]}_{\text{acceleration penalty}}, \quad \lambda(t) = (1-t)^2,$$

$$\mathbb{E} \left\| \frac{d^2 x_t}{dt^2} \right\|_2^2 \approx \frac{1}{\Delta t^2} \mathbb{E} \|u_\theta(t, x_t) - u_\theta(t + \Delta t, x_{t+\Delta t})\|_2^2.$$

This cannot be evaluated directly, because x_t and $x_{t+\Delta t}$ lie on the same *marginal* trajectory, which is unknown. FABO, the Flow Acceleration Upper Bound, uses $\tilde{x}_t$ and $\tilde{x}_{t+\Delta t}$ from the same *conditional* trajectory instead: they are easy to draw and need no knowledge of the marginal trajectory.

The FABO identity the equality behind the upper bound, from the last step of its proof (Appendix B); it holds for each fixed t

$$\underbrace{\mathbb{E} \|u_\theta(t, \tilde{x}_t) - u_\theta(t + \Delta t, \tilde{x}_{t+\Delta t})\|_2^2}_{\text{FABO evaluated on a conditional trajectory}} = \underbrace{\mathbb{E} \|u_\theta(t, x_t) - u_\theta(t + \Delta t, x_{t+\Delta t})\|_2^2}_{\text{1 Flow acceleration on the marginal trajectory}} + \underbrace{\Delta t^2 \mathbb{E} \left[\text{tr Var} \left[\frac{\partial u_\theta}{\partial x}(\tilde{x}_1 - \tilde{x}_0) \mid \tilde{x}_t \right] \right]}_{\text{2 Error term generated by FABO acts as a Jacobian regularizer}} + o(\Delta t^2)$$

1 Flow acceleration

The finite-difference form of the acceleration penalty in the objective, taken along the marginal trajectory.

- Smaller accelerations mean smoother, low-curvature flow trajectories. With zero acceleration the trajectory becomes a straight line.
- It cannot be evaluated directly: x_t and $x_{t+\Delta t}$ lie on the same marginal trajectory, which is unknown.
- The weight $\lambda(t) = (1-t)^2$ encourages smoother flow at earlier timesteps and prioritizes accuracy as $t \rightarrow 1$.

How the identity is obtained (Appendix B)

Assume u_θ is twice continuously differentiable with bounded second derivatives. Two trajectories pass through the same state $x_t = \tilde{x}_t$: the optimal marginal trajectory and the linear conditional trajectory $\tilde{x}_t = (1-t)\tilde{x}_0 + t\tilde{x}_1$.

1. Taylor expansion along the marginal trajectory, whose velocity is $u^{\text{opt}}(t, \tilde{x}_t) = \mathbb{E}[\tilde{x}_1 - \tilde{x}_0 \mid \tilde{x}_t]$ [3]:

$$\mathbb{E} \|u_\theta(t, x_t) - u_\theta(t + \Delta t, x_{t+\Delta t})\|_2^2 = \mathbb{E} \left\| \frac{\partial u_\theta}{\partial t} \Delta t + \frac{\partial u_\theta}{\partial x} \mathbb{E}[\tilde{x}_1 - \tilde{x}_0 \mid \tilde{x}_t] \Delta t + o(\Delta t) \right\|_2^2.$$

Effect in experiments

Average success (%), 12 MimicGen tasks, 100 demonstrations

EfficientFlow	52.6
Equivariant + MP1 [7]	46.3
Equivariant + Consistency FM [6]	40.4
No acceleration reg.	39.3

λ on Mug Cleanup D1	$0.5(1-t)^2$	$(1-t)^2$	$2(1-t)^2$	constant 0.5
Success rate (%)	48.0	50.0	51.3	42.0

Mean velocity change over 500 sampled timesteps on Stack D1: 0.103, against 0.136 without the regularization, a 24.3% reduction.

2 Error term: a Jacobian regularizer

- **Upper bound.** It is non-negative; when positive, $\text{FABO} \geq \text{1}$ for small Δt , so FABO minimizes an upper bound on the true acceleration penalty.
- **Size** (Appendix C). If $0 < \mu_1 \leq \text{Var}[a^\top(\tilde{x}_1 - \tilde{x}_0) \mid \tilde{x}_t] \leq \mu_2$ for all $\tilde{x}_t$ and unit vectors a ,

$$\mu_1 \mathbb{E} \left\| \frac{\partial u_\theta}{\partial x} \right\|_F^2 \leq \mathbb{E} \left[\text{tr Var} \left[\frac{\partial u_\theta}{\partial x}(\tilde{x}_1 - \tilde{x}_0) \mid \tilde{x}_t \right] \right] \leq \mu_2 \mathbb{E} \left\| \frac{\partial u_\theta}{\partial x} \right\|_F^2.$$
- **Role.** It penalizes the Jacobian $\partial u_\theta / \partial x$, so the velocity field becomes less sensitive, and therefore more robust, to perturbations of the state x_t .

2. Total expectation lets $\tilde{x}_1 - \tilde{x}_0$ replace its conditional mean in the cross term, and the squared terms differ by a conditional variance:

$$\mathbb{E} \left[\left\| \frac{\partial u_\theta}{\partial x}(\tilde{x}_1 - \tilde{x}_0) \Delta t \right\|_2^2 - \left\| \frac{\partial u_\theta}{\partial x} \mathbb{E}[\tilde{x}_1 - \tilde{x}_0 \mid \tilde{x}_t] \Delta t \right\|_2^2 \right] = \mathbb{E} \left[\text{tr Var} \left[\frac{\partial u_\theta}{\partial x}(\tilde{x}_1 - \tilde{x}_0) \Delta t \mid \tilde{x}_t \right] \right].$$

3. The rest is the expansion of the conditional difference:

$$\mathbb{E} \left\| \frac{\partial u_\theta}{\partial t} \Delta t + \frac{\partial u_\theta}{\partial x}(\tilde{x}_1 - \tilde{x}_0) \Delta t + o(\Delta t) \right\|_2^2 = \mathbb{E} \|u_\theta(t, \tilde{x}_t) - u_\theta(t + \Delta t, \tilde{x}_{t+\Delta t})\|_2^2 + o(\Delta t^2).$$

Average over 12 tasks

Method	NFE	Runtime (ms)	Success (%) by number of demos		
			100	200	1000
Ours	1	12.22	52.61	66.18	75.25
Ours	3	22.59	53.49	69.33	81.36
Ours	5	34.45	54.18	70.26	81.99
EquiDiff	100	685.92	53.77	68.59	79.69
DP-C	100	542.96	42.00	57.75	71.42
DP-T	100	497.53	29.00	43.00	64.92
DP3	10	53.83	23.92	35.08	56.75
ACT	1	12.51	21.33	38.17	63.25

- With 5 NFE, above EquiDiff at every data size and **19.9×** faster.
- 200 demonstrations reach 98.4% of the success DP-C obtains with 1000 (5 NFE).

Epochs to reach 50% of peak success

EfficientFlow	31.7
No accel. reg.	35.0
EquiDiff	51.7

Mean over 12 tasks, 100 demos, evaluated every 10 epochs. On Hammer Cleanup: 10 epochs against 50 for EquiDiff.

Voxel input MimicGen, 100 demos

Method	Stack	Thread.	Square	Stack 3	3-Pc.	Avg.
DP3 [8]	69.3	12.0	6.7	7.3	0.0	19.1
FlowPolicy [10]	72.0	13.3	6.0	10.0	0.0	20.3
Ours (RGB)	94.0	31.3	20.7	48.0	10.7	41.0
Ours (Voxel)	93.3	41.3	33.3	67.3	20.0	51.0

Robomimic [11] 20 demos

Method	Tool Hang	Can	Lift	Square	Avg.
DP-C [5]	15.3	67.3	100.0	42.7	56.3
Ours	16.7	90.7	100.0	44.0	62.9

Takeaways

Equivariance carries over to flow matching with an isotropic prior and an equivariant velocity network.

FABO is the flow acceleration plus a Jacobian regularizer: one tractable loss that needs only conditional trajectories.

One-step inference at 81.8 Hz, 56.1× faster than EquiDiff, with competitive success.